Solving (Optimization) problems: differentiation

✓ optmzn. problems :- PCA, logistic regrsn ; lr. regression

✓ ML :- { differentiation → single-value, vector
✓ maxima & minima → 11th & 12th → { 99% }
→ Undergrad

Integration
Diff. eqns { less used }

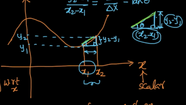
Single-variable diffn: $y = f(x)$

$y = f(x)$

$\left\{ \frac{dy}{dx} = \frac{df}{dx} = y' = f' \right\}$

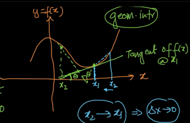
↳ differentiation of y w.r.t x

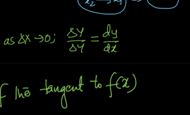
$\frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta Y}{\Delta X} = \tan\theta$


y_2
y_1
$y_2 - y_1$
$(y_2 - y_1)$
$(x_2 - x_1)$
θ
x_1
x_2
x
scalar

{ "$\frac{dy}{dx}$" → intuitively → rate of change of y as x changes
→ how much does y change as x changes

$$\left[\frac{dy}{dx}\right] = \lim_{\Delta x \to 0} \frac{\Delta Y}{\Delta X}$$




$y = f(x)$
geom. intr
Tangent of f(x) @ x_1
θ
θ'
x
x_2
x_1
x_2

Tangent is the hyp that we obtain as $\delta x \to 0$

$x_2 \to x_1 \Rightarrow \delta x \to 0$

$$\boxed{\tan\theta = \frac{dy}{dx}} \quad \text{as } \delta x \to 0; \ \frac{\delta y}{\delta y} = \frac{dy}{dx}$$

$\frac{dy}{dx}$ = Slope of the tangent to f(x)

$\left[\frac{dy}{dx}\right]_{x_1}$ = Slope of the tangent ~~of~~ to f(x) @ $x = x_1$

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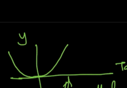
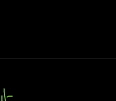
differentiation :- Late 1600's
↳ Newton ✓ → Principia Mathematica ✓
↳ Leibnitz ✓ → ∫ ①

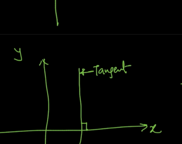
disp ; velocity ; acc ; gravity ; orbital
↓
rate of change

$\tan\theta$ = slope
↳ Tangent & x-axis

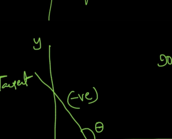

y
Tangent
θ
x

$0 \leq \theta < 90$:- +ve

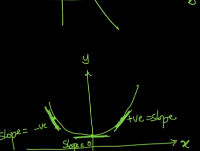



y
Tangent
parallel
x

$\tan\theta = 0$

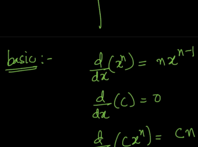


Tan 90 : undefined.


y
Tangent
(-ve)
θ
x

$90 < \theta < 180$

$\tan\theta$: -ve


y
slope = -ve
+ve = slope
Slope = 0
x

slope = $\tan\theta$

geom

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basic :-

$\frac{d}{dx}(x^n) = n x^{n-1}$; $\frac{d}{dx}(x^2) = 2x$

$\frac{d}{dx}(3x) = 3$

$\frac{d}{dx}(c) = 0$

$\frac{d}{dx}(c x^n) = c n x^{n-1}$

$\frac{d}{dx}\log(x) = 1/x$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx}\left(f(x) \overset{-}{+} g(x)\right) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

Chain-rule : $\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx}$

Jeff. dean

$f(g(x)) = (a - bx)^2$

$g(x) = (a - bx) = z$

$f(x) = x^2$

✓ $\frac{df(g(x))}{dx} = -2b(a - bx)$

$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \frac{dg}{dx}$

$\frac{df}{dg} \to 2(a - bx)$; $\frac{dg}{dx} \to (-b)$

$\frac{dg}{dx} = \frac{d}{dx}(a - bx) = \cancel{\frac{d}{dx}(a)} - \frac{d}{dx}(bx) = -b$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx}\left(f(x) \overset{-}{+} g(x)\right) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$